

README

Here is a comprehensive summary of your **Bonus-B Enterprise Infrastructure** deployment. This is formatted to be "copy-paste ready" for your lab report or GitHub README.

Infrastructure as Code: Final Validation Report

Project Name: `lew` (Chewbacca Lab)

Status:  Fully Deployed & Validated

Region: `ap-northeast-1` (Tokyo)

1. Network & Connectivity

- **DNS Authority:** Successfully migrated from GoDaddy to **AWS Route 53**.
- **Load Balancer:** `lew-alb` is in an `active` state.
- **Listeners:** * **Port 80:** Configured for automatic HTTP to HTTPS redirection.
 - **Port 443:** Secure HTTPS listener active with ACM SSL termination.
- **Domain Resolution:** `projectcomplete.me` nameservers verified via `dig`.

2. Security & Identity

- **SSL Certificate:** ACM Certificate for `projectcomplete.me` is **Issued** and attached to the ALB.
- **WAF (Web Application Firewall):** `lew-waf` is associated with the ALB resource, providing protection against common web exploits (SQLi, XSS).

3. Compute & Health

- **Target Group:** `lew-tg01` is managing backend EC2 instances.
- **Health Status:** Verified `healthy`. The ALB is successfully receiving 200 OK responses from the application layer.

4. Observability (Bonus-B Requirements)

- **CloudWatch Alarms:** `lew-alb-5xx-alarm01` is active. It monitors the Load Balancer for server-side errors, ensuring high availability.
- **CloudWatch Dashboard:** `lew-dashboard01` provides a centralized view of:
 - Request Count
 - Target Response Time
 - 5xx / 4xx Error rates

- WAF Blocked Requests

Final CLI Verification Log (Condensed)

Resource	Command	Result
ALB State	<code>aws elbv2 describe-load-balancers</code>	<code>active</code>
SSL Listener	<code>aws elbv2 describe-listeners</code>	<code>Port: 443</code>
Target Health	<code>aws elbv2 describe-target-health</code>	<code>healthy</code>
WAF Config	<code>aws wafv2 get-web-acl-for-resource</code>	<code>lew-waf</code>
Observability	<code>aws cloudwatch list-dashboards</code>	<code>lew-dashboard 01</code>

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FINAL VERIFICATION LOG: BONUS-B INFRASTRUCTURE

1. Load Balancer ARN Discovery Command: `aws elbv2 describe-load-balancers --names "lew-alb" --query "LoadBalancers[0].LoadBalancerArn" --output text`

Result:

```
lab-1c-chewbacca on  main [x!?] via  default on  (ap-northeast-1) on  took 2s
> aws elbv2 describe-load-balancers --names "lew-alb" --query "LoadBalancers[0].LoadBalancerArn" --output text
arn:aws:elasticloadbalancing:ap-northeast-1:127214194163:loadbalancer/app/lew-alb/8c527aae6d8d5158
```

arn:aws:elasticloadbalancing:ap-northeast-1:127214194163:loadbalancer/app/lew-alb/8c527aae6d8d5158

2. ALB State Verification Command: `aws elbv2 describe-load-balancers --load-balancer-arns arn:aws:elasticloadbalancing:ap-northeast-1:127214194163:loadbalancer/app/lew-alb/8c527aae6d8d5158 --query "LoadBalancers[0].State.Code" --output text`

Result:

```
lab-1c-chewbacca on  main [x!?] via  default on  (ap-northeast-1) on 
> aws elbv2 describe-load-balancers --load-balancer-arns arn:aws:elasticloadbalancing:ap-northeast-1:127214194163:loadbalancer/app/lew-alb/8c527aae6d8d5158 --query "LoadBalancers[0].State.Code" --output text
active
```

active

3. HTTPS Listener Verification (Port 443) Command: `aws elbv2 describe-listeners --load-balancer-arn arn:aws:elasticloadbalancing:ap-northeast-1:127214194163:loadbalancer/app/lew-alb/8c527aae6d8d5158 --query "Listeners[0].Port" --output text`

Result:

```
lab-1c-chewbacca on  main [x!?] via  default on  (ap-northeast-1) on 
> ALB_ARN=$(terraform state show aws_lb.chewbacca_alb01 | grep "^arn " | cut -d'=' -f2 | tr -d ' ')
aws elbv2 describe-listeners --load-balancer-arn $ALB_ARN --query "Listeners[0].Port" --output text
80      443
```

80 443

4. Target Group Health Check Command: `aws elbv2 describe-target-health --target-group-arn arn:aws:elasticloadbalancing:ap-northeast-1:127214194163:targetgroup/lew-tg/1dd46ad835f8f6af --query "TargetHealthDescriptions[].TargetHealth.State" --output text`

Result:

```
lab-1c-chewbacca on  main [x!?] via  default on  (ap-northeast-1) on 
> TG_ARN=$(terraform output -raw target_group_arn 2>/dev/null || terraform state show aws_lb_target_group.chewbacca_tg01 | grep "^    arn " | cut -d'=' -f2 | tr -d ' ')

aws elbv2 describe-target-health --target-group-arn $TG_ARN --query "TargetHealthDescriptions[].TargetHealth.State" --output text

healthy
```

healthy

5. WAF (Web Application Firewall) Attachment Command: `aws wafv2 get-web-acl-for-resource --resource-arn arn:aws:elasticloadbalancing:ap-northeast-1:127214194163:loadbalancer/app/lew-alb/8c527aae6d8d5158 --query "WebACL.Name" --output text`

Result:

```
lab-1c-chewbacca on  main [x!?] via  default on  (ap-northeast-1) on  took 8s
● > aws wafv2 get-web-acl-for-resource --resource-arn $ALB_ARN --query "WebACL.Name" --output text
lew-waf
```

lew-waf

6. CloudWatch Alarm Status Command: `aws cloudwatch describe-alarms --alarm-names "lew-alb-5xx-alarm01" --query "MetricAlarms[0].StateValue" --output text`

Result:

```
lab-1c-chewbacca on  main [x!?] via  default on  (ap-northeast-1) on 
● > aws cloudwatch describe-alarms \
  --alarm-names "lew-alb-5xx-alarm01" \
  --query "MetricAlarms[0].StateValue" \
  --output text
INSUFFICIENT_DATA
```

INSUFFICIENT_DATA

7. CloudWatch Dashboard Presence Command: `aws cloudwatch list-dashboards --dashboard-name-prefix "lew" --query "DashboardEntries[0].DashboardName" --output text`

Result:

```
lab-1c-chewbacca on  main [x!?] via  default on  (ap-northeast-1) on 
● > aws cloudwatch list-dashboards \
  --dashboard-name-prefix "lew" \
  --query "DashboardEntries[0].DashboardName" \
  --output text
lew-dashboard01
```

lew-dashboard01

8. DNS & Load Balancer Endpoint Command: `aws elbv2 describe-load-balancers --names "lew-alb" --query "LoadBalancers[0].DNSName" --output text`

Result:

```
lab-1c-chewbacca on  main [x!?] via  default on  (ap-northeast-1) on  took 2s
● > aws elbv2 describe-load-balancers \
  --names "lew-alb" \
  --query "LoadBalancers[0].DNSName" \
  --output text
lew-alb-2104103463.ap-northeast-1.elb.amazonaws.com
```

lew-alb-2104103463.ap-northeast-1.elb.amazonaws.com

9. External Connectivity Test Command: `url -I --resolve projectcomplete.me:443:$(dig +short lew-alb-2104103463.ap-northeast-1.elb.amazonaws.com | tail -n1) https://projectcomplete.me`

Result: curl:

```
lab-1c-chewbacca on  main [x!?] via  default on  (ap-northeast-1) on 
● > curl -I --resolve projectcomplete.me:443:$(dig +short lew-alb-2104103463.ap-northeast-1.elb.amazonaws.com | tail -n1) https://projectcomplete.me
HTTP/2 200
date: Sun, 08 Feb 2026 20:47:48 GMT
content-type: text/html; charset=utf-8
content-length: 93
server: Werkzeug/3.1.5 Python/3.9.25
```

HTTP/2 200
date: Sun, 08 Feb 2026 20:47:48 GMT
content-type: text/html; charset=utf-8
content-length: 93
server: Werkzeug/3.1.5 Python/3.9.25